Rapidity Problems Part 1 Physics 7502 Spring 2026

R1) Show that

$p_{||} = m_T \sinh y$

$E = m_T \cosh y$

These previously had cosh/sinh in the wrong order rather than the correct way as now shown, w/ sinh in p prelation, E with cosh

where $m_T = \sqrt{m^2 + p_T^2}$+

Use definitions of the hyperbolic functions in terms of exponentials.

R2) Show the following relations

a) $d\eta = d\theta / \sin\theta$ (another way to prove the equivalence of the two differentials at y = 0—start with a different form of $\eta(\theta)$ than the one on the slide that was used for this purpose)

b) $\sin\theta = 1 / \cosh \eta$ when m= 0

c) $dp_{||} / E$ is invariant

R3) Show that dy is invariant in a different way than we did in the slides (ie not taking the limit of Δy)

Hint : use 2 c)